

1. Stack levantado y todos los servicios accesibles

```
PS C:\7MO CICLO\INFRA\S10\iac-observabilidad> docker compose up -d --build
#18 exporting manifest list sha256:de26f9021006e58ad3bfb039beec5def24e21e63ef733a591d883df11ada49b 0.0s done
#18 naming to docker.io/library/iac-observabilidad-backend:latest done
#18 unpacking to docker.io/library/iac-observabilidad-backend:latest done
#18 DONE 0.1s

#19 [backend] resolving provenance for metadata file
#19 DONE 0.0s

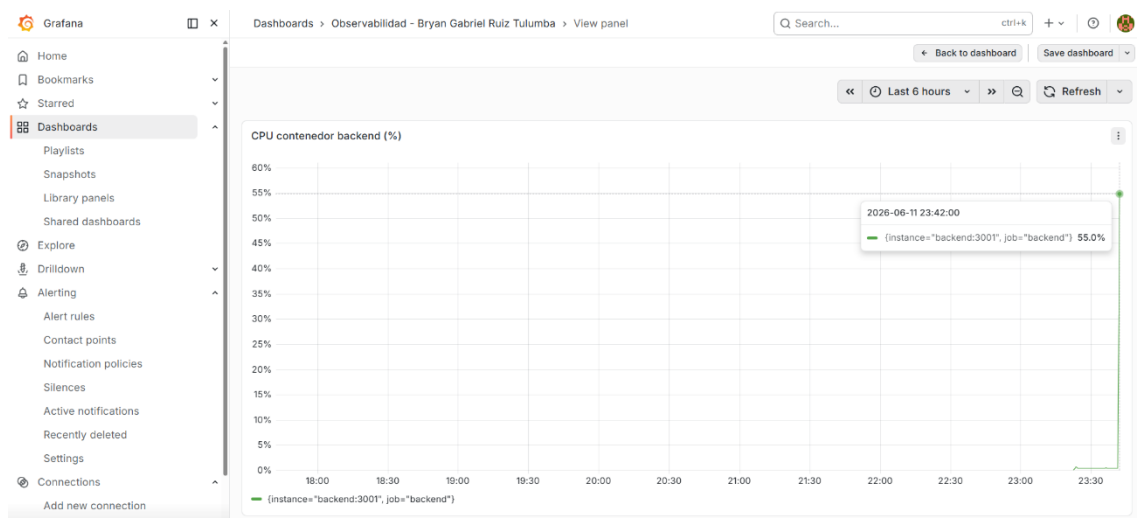
#20 [frontend] resolving provenance for metadata file
#20 DONE 0.0s
[+] up 13/13
✓ Image iac-observabilidad-backend          Built
✓ Image iac-observabilidad-frontend         Built
✓ Network iac-observabilidad_default        Created
✓ Volume iac-observabilidad_prometheus-data Created
✓ Volume iac-observabilidad_grafana-data    Created
✓ Container lab-prometheus                  Started
✓ Container lab-cadvisor                    Started
✓ Container lab-loki                        Started
✓ Container lab-backend                     Started
✓ Container lab-node-exporter               Started
✓ Container lab-frontend                    Started
... 2 more
```

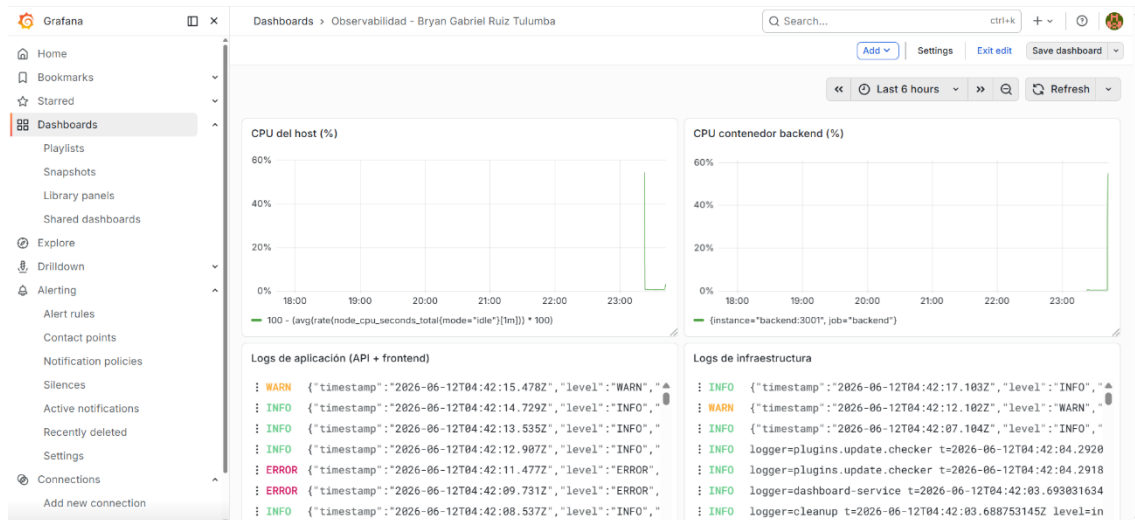
```
PS C:\7MO CICLO\INFRA\S10\iac-observabilidad> docker compose ps
```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS
lab-alloy	grafana/alloy:v1.10.0	"/bin/alloy run --se..."	alloy	12 seconds ago	Up 10 seconds
lab-backend	iac-observabilidad-backend	"docker-entrypoint.s..."	backend	12 seconds ago	Up 11 seconds
lab-cadvisor	gcr.io/cadvisor/cadvisor:v0.49.1	"/usr/bin/cadvisor -..."	cadvisor	12 seconds ago	Up 11 seconds (h
lab-frontend	iac-observabilidad-frontend	"docker-entrypoint.s..."	frontend	12 seconds ago	Up 10 seconds
lab-grafana	grafana/grafana:12.4.0	"/run.sh"	grafana	12 seconds ago	Up 10 seconds
lab-loki	grafana/loki:3.4.0	"/usr/bin/loki --con..."	loki	12 seconds ago	Up 11 seconds
lab-node-exporter	prom/node-exporter:v1.8.2	"/bin/node_exporter ..."	node-exporter	12 seconds ago	Up 11 seconds
lab-prometheus	prom/prometheus:latest	"/bin/prometheus --c..."	prometheus	12 seconds ago	Up 11 seconds

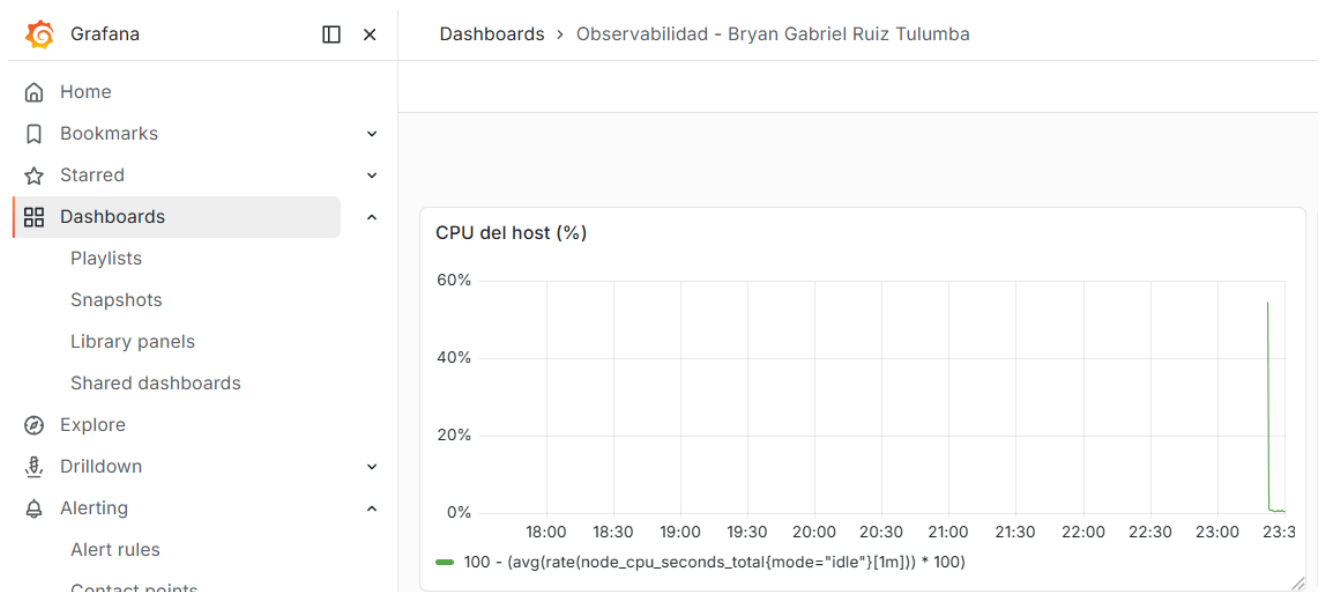
```
PS C:\7MO CICLO\INFRA\S10\iac-observabilidad>
```

2. Dashboard con panel de CPU por contenedor (con umbral en 50%)





3. Dashboard con panel de CPU del host



4. Panel de logs de aplicación funcionando (filtro por nivel) INFO

Grafana Dashboard: Observabilidad - Bryan Gabriel Ruiz Tulumba

Search: Search... (ctrl+k)

Buttons: Add, Settings, Exit edit, Save dashboard

Time Range: Last 6 hours

Refresh: Refresh

Logs de aplicación (API + frontend)

```
INFO {"timestamp":"2026-06-12T05:41:15.376Z","level":"INFO","service":"backend","msg":"pedido_creado","order_id":7125,"amount":146}
INFO {"timestamp":"2026-06-12T05:41:14.579Z","level":"INFO","service":"frontend","msg":"click_boton","button":"cargar"}
INFO {"timestamp":"2026-06-12T05:41:13.284Z","level":"INFO","service":"backend","msg":"http_request","method":"GET","path":"/metrics"}
INFO {"timestamp":"2026-06-12T05:41:12.657Z","level":"INFO","service":"frontend","msg":"http_request","method":"GET","path":"/metric"}
INFO {"timestamp":"2026-06-12T05:41:11.378Z","level":"INFO","service":"backend","msg":"pedido_creado","order_id":1573,"amount":345}
INFO {"timestamp":"2026-06-12T05:41:09.581Z","level":"INFO","service":"frontend","msg":"click_boton","button":"saludar"}
INFO {"timestamp":"2026-06-12T05:41:08.289Z","level":"INFO","service":"backend","msg":"http_request","method":"GET","path":"/metrics"}
INFO {"timestamp":"2026-06-12T05:41:07.657Z","level":"INFO","service":"frontend","msg":"http_request","method":"GET","path":"/metric"}
INFO {"timestamp":"2026-06-12T05:41:03.378Z","level":"INFO","service":"backend","msg":"usuario_autenticado","user_id":5}
INFO {"timestamp":"2026-06-12T05:41:03.289Z","level":"INFO","service":"backend","msg":"http_request","method":"GET","path":"/metrics"}
INFO {"timestamp":"2026-06-12T05:41:02.657Z","level":"INFO","service":"frontend","msg":"http_request","method":"GET","path":"/metric"}
INFO {"timestamp":"2026-06-12T05:40:59.581Z","level":"INFO","service":"frontend","msg":"click_boton","button":"saludar"}
INFO {"timestamp":"2026-06-12T05:40:58.289Z","level":"INFO","service":"backend","msg":"http_request","method":"GET","path":"/metrics"}
INFO {"timestamp":"2026-06-12T05:40:57.657Z","level":"INFO","service":"frontend","msg":"http_request","method":"GET","path":"/metric"}
INFO {"timestamp":"2026-06-12T05:40:55.377Z","level":"INFO","service":"backend","msg":"pago_procesado","order_id":9885,"gateway":"st"}
```

WARN

Grafana Dashboard: Observabilidad - Bryan Gabriel Ruiz Tulumba

Search: Search... (ctrl+k)

Buttons: Add, Settings, Exit edit, Save dashboard

Time Range: Last 6 hours

Refresh: Refresh

Logs de aplicación (API + frontend)

```
WARN {"timestamp":"2026-06-12T05:41:04.588Z","level":"WARN","service":"frontend","msg":"recurso_lento","asset":"main.js","load_ms":1}
WARN {"timestamp":"2026-06-12T05:40:54.581Z","level":"WARN","service":"frontend","msg":"recurso_lento","asset":"main.js","load_ms":2}
WARN {"timestamp":"2026-06-12T05:40:24.582Z","level":"WARN","service":"frontend","msg":"recurso_lento","asset":"main.js","load_ms":1}
WARN {"timestamp":"2026-06-12T05:40:20.638Z","level":"WARN","service":"frontend","msg":"carga_cpu_solicitada_desde_frontend"}
WARN {"timestamp":"2026-06-12T05:40:20.637Z","level":"WARN","service":"backend","msg":"cpu_load_test_started","seconds":30}
WARN {"timestamp":"2026-06-12T05:40:07.380Z","level":"WARN","service":"backend","msg":"reintento_de_pago","order_id":5991,"attempt":1}
WARN {"timestamp":"2026-06-12T05:39:39.583Z","level":"WARN","service":"frontend","msg":"recurso_lento","asset":"main.js","load_ms":2}
WARN {"timestamp":"2026-06-12T05:39:14.583Z","level":"WARN","service":"frontend","msg":"recurso_lento","asset":"main.js","load_ms":1}
WARN {"timestamp":"2026-06-12T05:39:11.380Z","level":"WARN","service":"backend","msg":"latencia_alta_en_db","query_ms":1425}
WARN {"timestamp":"2026-06-12T05:38:55.380Z","level":"WARN","service":"backend","msg":"reintento_de_pago","order_id":1534,"attempt":1}
WARN {"timestamp":"2026-06-12T05:38:43.379Z","level":"WARN","service":"backend","msg":"reintento_de_pago","order_id":2020,"attempt":1}
WARN {"timestamp":"2026-06-12T05:38:31.382Z","level":"WARN","service":"backend","msg":"reintento_de_pago","order_id":5601,"attempt":1}
WARN {"timestamp":"2026-06-12T05:38:27.382Z","level":"WARN","service":"backend","msg":"reintento_de_pago","order_id":7988,"attempt":1}
WARN {"timestamp":"2026-06-12T05:38:19.584Z","level":"WARN","service":"frontend","msg":"recurso_lento","asset":"main.js","load_ms":1}
WARN {"timestamp":"2026-06-12T05:38:15.382Z","level":"WARN","service":"backend","msg":"latencia_alta_en_db","query_ms":2160}
```

CPU contenedor backend (%)

Logs de infraestructura

ERROR

Grafana Dashboard: Observabilidad - Bryan Gabriel Ruiz Tulumba

Search: Search... (ctrl+k)

Buttons: Add, Settings, Exit edit, Save dashboard

Time Range: Last 6 hours

Refresh: Refresh

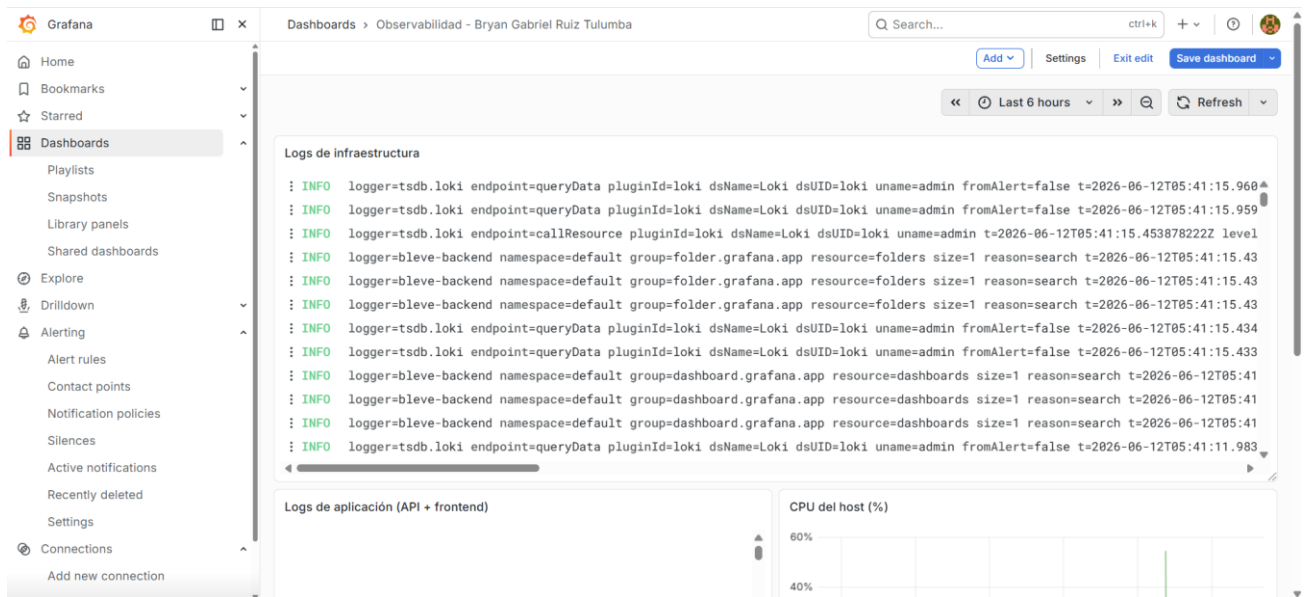
Logs de aplicación (API + frontend)

```
ERROR {"timestamp":"2026-06-12T05:41:07.378Z","level":"ERROR","service":"backend","msg":"excepcion_no_controlada","trace_id":"c57500"}
ERROR {"timestamp":"2026-06-12T05:40:59.377Z","level":"ERROR","service":"inventory","msg":"fallo_conexion_inventario","code":503}
ERROR {"timestamp":"2026-06-12T05:40:51.378Z","level":"ERROR","service":"backend","msg":"excepcion_no_controlada","trace_id":"47a352"}
ERROR {"timestamp":"2026-06-12T05:40:44.580Z","level":"ERROR","service":"frontend","msg":"error_js_cliente","message":"TypeError: un"}
ERROR {"timestamp":"2026-06-12T05:40:27.379Z","level":"ERROR","service":"inventory","msg":"fallo_conexion_inventario","code":503}
ERROR {"timestamp":"2026-06-12T05:40:19.582Z","level":"ERROR","service":"frontend","msg":"error_js_cliente","message":"TypeError: un"}
ERROR {"timestamp":"2026-06-12T05:40:19.379Z","level":"ERROR","service":"backend","msg":"excepcion_no_controlada","trace_id":"ab1832"}
ERROR {"timestamp":"2026-06-12T05:40:14.582Z","level":"ERROR","service":"frontend","msg":"error_js_cliente","message":"TypeError: un"}
ERROR {"timestamp":"2026-06-12T05:40:11.380Z","level":"ERROR","service":"backend","msg":"excepcion_no_controlada","trace_id":"b643bd"}
ERROR {"timestamp":"2026-06-12T05:40:03.379Z","level":"ERROR","service":"inventory","msg":"fallo_conexion_inventario","code":503}
ERROR {"timestamp":"2026-06-12T05:39:59.582Z","level":"ERROR","service":"frontend","msg":"error_js_cliente","message":"TypeError: un"}
ERROR {"timestamp":"2026-06-12T05:39:59.379Z","level":"ERROR","service":"backend","msg":"excepcion_no_controlada","trace_id":"f7faf6"}
ERROR {"timestamp":"2026-06-12T05:39:55.379Z","level":"ERROR","service":"backend","msg":"excepcion_no_controlada","trace_id":"23d1f0"}
ERROR {"timestamp":"2026-06-12T05:39:51.379Z","level":"ERROR","service":"backend","msg":"excepcion_no_controlada","trace_id":"fc54a3"}
ERROR {"timestamp":"2026-06-12T05:39:44.581Z","level":"ERROR","service":"frontend","msg":"error_js_cliente","message":"TypeError: un"}
```

CPU contenedor backend (%)

Logs de infraestructura

5. Panel de logs de infraestructura funcionando



6. Alarma de CPU > 50% configurada correctamente

The screenshot shows the Grafana 'Alerting' configuration steps for a new alert rule. The steps are as follows:

- 1. Enter alert rule name**
Enter a name to identify your alert rule.
Name: CPU backend > 50%
- 2. Define query and alert condition**
Define query and alert condition. [Need help?](#)
Prometheus [Options](#) 10m to now, MD = 43200, Min. Interval = 1s
Explain [Metrics browser](#) [Run queries](#) [Builder](#) [Code](#)
Query: `rate(backend_process_cpu_seconds_total[1m]) * 100`
Options Legend: Auto Format: Time series Step: auto Type: Instant
Table

Query	Value
{instance="backend:3001", job="backend"}	0.46133

Alert condition
WHEN QUERY IS ABOVE 50
{instance="backend:3001", job="backend"} 0 Normal
[Preview alert rule condition](#)
- 3. Add folder and labels**
Organize your alert rule with a folder and set of labels. [Need help?](#)
Folder
Select a folder to store your rule in.
alerta [+ New folder](#)
Labels
Add labels to your rule for searching, silencing, or routing to a notification policy. [Need help?](#)
severity: warning [Edit labels](#)
- 4. Set evaluation behavior**
Define how the alert rule is evaluated. [Need help?](#)
Evaluation group and interval
alertas or [+ New evaluation group](#)
All rules in the selected group are evaluated every 10s.
Pending period
Period during which the threshold condition must be met to trigger an alert. Selecting "None" triggers the alert immediately once the condition is met.
30s
None 10s 20s 30s 40s 50s
Keep firing for
Period during which the alert will continue to show up as firing even though the threshold condition is no longer breached. Selecting "None" means the alert will be back to normal immediately.
0s
None 10s 20s 30s 40s 50s

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New alert rule

Q Search...

ctrl+k

+

+

5. Configure notifications

Select who should receive a notification when an alert rule fires.

Advanced options

Recipient

Notifications for firing alerts are routed to a selected contact point. [Need help?](#)

Alertmanager: grafana

Contact point

empty

[View or create contact points](#)

Muting, grouping and timings (optional)

6. Configure notification message

Add more context to your alert notifications. [Need help?](#)

Summary (optional)

Short summary of what happened and why.

Enter a summary...

Description (optional)

Description of what the alert rule does.

Enter a description...

Runbook URL (optional)

Webpage where you keep your runbook for the alert.

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alertra > alertas > CPU backend > 50%

Q Search...

ctrl+k

+

+

CPU backend > 50%

Normal

Notifications are delivered to empty

Evaluation interval Every 10s

Labels severity warning

Edit

More

Query and conditions

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A Prometheus 10m to now

[View in Explore](#)

rate[back_end_process_cpu_seconds_total[1m]] * 100

Table

{instance="backend:3001", job="backend"}

0.49349

C Threshold

[Alert condition](#)

Input A Is above 50

{instance="backend:3001", job="backend"}

0 Normal

7. Evidencia de la alarma en estado *Firing* tras generar carga EN PENDING

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ctrl+k

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Alert rules

Rules that determine whether an alert will fire

New alert rule

Export rule definition

Search by data sources

Dashboard

All data sources

Select dashboard

State

Rule type

Health

Contact point

Firing

Normal

Pending

Recovering

Alert

Recording

Ok

No Data

Error

Choose a contact point

Search

View as

Grouped

List

State

1 rule 1 pending

Grafana-managed

Export rules

New recording rule

alertra > alertas

1 pending

10s

State

Name

Health

Summary

Next evaluation

Actions

Pending for 8s

CPU backend > 50%

ok

in a few seconds

View

Edit

More

RESULTADO EN FIRING

The screenshot shows the Grafana Alerting 'Alert rules' page. A sidebar on the left contains navigation links like Home, Bookmarks, Starred, Dashboards, Playlists, Snapshots, Library panels, Shared dashboards, Explore, Drilldown, Alerting, and Connections. The main area is titled 'Alert rules' and shows a list of rules. One rule, 'alerta', is in a 'Firing' state. Below the list, a table shows the details of the firing rule:

State	Name	Health	Summary	Next evaluation	Actions
Firing	for 9s CPU backend > 50%	ok		in a few seconds	View Edit More

8. Ciclo cerrado alarma → log vía webhook

The screenshot shows a Grafana dashboard titled 'Observabilidad - Bryan Gabriel Ruiz Tulumba'. It displays a table of logs from the application (API + frontend). The logs show three error messages, all with the message 'grafana_alert_received' and 'alert_status':

```
{ "timestamp": "2026-06-12T05:28:18.391Z", "level": "ERROR", "service": "backend", "msg": "grafana_alert_received", "alert_status": "Firing" }
{ "timestamp": "2026-06-12T05:28:15.119Z", "level": "ERROR", "service": "backend", "msg": "grafana_alert_received", "alert_status": "Firing" }
{ "timestamp": "2026-06-12T05:07:18.177Z", "level": "ERROR", "service": "backend", "msg": "grafana_alert_received", "alert_status": "Firing" }
```

Breve explicación (5–8 líneas) de qué hace cada componente del stack.

Prometheus es el motor de las métricas del stack. Hace scraping periódico a los endpoints de cada servicio y los almacena. Permite consultarlas con PromQL.

Node Exporter expone las métricas del host (CPU, memoria, disco) en formato Prometheus. Y necesita acceso al sistema de archivos del host

cAdvisor hace lo mismo que node-exporter pero permite ver cuántos recursos consume cada contenedor Docker por separado.

Loki es como Prometheus pero para logs de cada contenedor. Indexa solo sus etiquetas lo que lo hace muy liviano

Grafana Alloy es el recolector de logs, captura el stdout de cada contenedor y los reenvía a Loki con etiquetas como tier="application" o tier="infrastructure".

Grafana se conecta a Prometheus y Loki como fuentes de datos y permite construir dashboards con métricas y logs. También gestiona las alarmas.

Backend y Frontend son las aplicaciones que usamos en el lab. Exponen métricas propias y emiten logs en formato JSON, siendo los sujetos de observación del stack.

LINK PROYECTO DE IAC:

https://github.com/Estalin7/proyecto_infra